

The Volume Dividend

Energy density, blend dilution, and who profits per extra litre — ethanol and isobutanol in petrol, biodiesel in diesel, versus compressed biogas in CNG

India, FY 2024-25 volume basis · July 2026 prices · companion to the OMC retail-profitability model

Executive summary

Ethanol carries 34% less energy per litre than petrol. Blending it dilutes every litre sold, cutting km-per-litre — yet the pump price per litre is unchanged, so the consumer silently buys more litres for the same distance. Because central excise, state VAT (in effect), dealer commission and OMC marketing margin are all levied per litre, every stakeholder in the fuel chain collects on the extra volume. At E20 on India's 54-billion-litre petrol pool, the volume effect alone moves ₹22,700 crore per year of extra consumer spend: ₹8,500 crore to the exchequer, ₹886 crore to retail-outlet dealers, ₹757 crore to OMC marketing margin. Walking the blend to E30 roughly doubles this.

Compressed biogas (CBG) is the structural exception: purified to IS 16087 spec it matches or exceeds pipeline CNG in energy per kilogram, and CNG is retailed per kg — so blending CBG into the gas stream substitutes imports without any hidden dilution levy on the consumer. It is, fiscally, the honest blend.

1. Energy content of the fuels (lower heating value)

Fuel	MJ/kg	MJ/litre	Sold by	vs base
Petrol (E0)	43.4	32.1	litre	—
Ethanol (anhydrous)	26.8	21.1	litre	−34%
Isobutanol	33.1	26.5	litre	−17%
Diesel (B0)	43.0	35.7	litre	—
Biodiesel (FAME)	37.2	32.7	litre	−8%
HVO renewable diesel	44.0	34.3	litre	−4% (drop-in)
CNG (pipeline gas)	47.5	n/a (gas)	kg	—
CBG (IS 16087, CH ₄ ≥90%)	46.5	n/a (gas)	kg	≈ CNG parity
Pure methane (CBG ceiling)	50.0	n/a (gas)	kg	+5% vs CNG

The oxygenates that policy favours for petrol are exactly the ones with the deepest energy deficit. Isobutanol — denser, less oxygenated — halves ethanol's penalty per unit of renewable content. On the gas side the ordering flips: well-purified biogas sits at or above fossil CNG.

2. Blend dilution → extra litres for the same distance

Blend	MJ/L	Energy vs base	Extra litres (energy basis)	Real-world drop
E10	31.02	−3.4%	+3.5%	~2%
E20 (today)	29.92	−6.8%	+7.3%	4% (SIAM 2–6%)
E25	29.37	−8.5%	+9.3%	~5.5%
E30	28.82	−10.3%	+11.4%	~7%
IB16 isobutanol	31.22	−2.8%	+2.9%	no India data
IB24 isobutanol	30.78	−4.2%	+4.3%	no India data

Blend	MJ/L	Energy vs base	Extra litres (energy basis)	Real-world drop
B7 biodiesel	35.48	-0.6%	+0.6%	≈ energy basis
B10 biodiesel	35.39	-0.8%	+0.8%	≈ energy basis
B15 biodiesel	35.25	-1.2%	+1.3%	≈ energy basis
B20 biodiesel	35.10	-1.7%	+1.7%	≈ energy basis
CBG in CNG (any %)	per kg	0%	0%	0% — sold per kg

Real-world E20 loss (4%) runs below the pure energy math (6.8%) because E20-calibrated engines claw back part of it through octane and combustion gains; older vehicles lose up to ~12%. Diesel has no equivalent recovery mechanism, so biodiesel's real-world drop tracks the energy arithmetic.

3. What the dilution costs a petrol owner

Hatchback, 20 km/L on E0, 10,000 km/yr, pump ₹105/L. The price board shows the same ₹/litre for E0 and E20 — the energy cut is invisible at the point of sale:

Blend	Extra litres / yr	Extra ₹ / yr	Hidden levy on every km
E20	+21 L	₹2,188	4.2%
E25	+29 L	₹3,056	5.8%
E30	+38 L	₹3,952	7.5%

4. National petrol pool: where the extra litres' money goes

Base: FY24-25 petrol pool 54.1 bn L (40 MMT), distance demand held at its E0-equivalent, dispensed through 99,281 retail outlets. Per-litre stakes: excise ₹19.90, effective state VAT ≈₹19.5, dealer commission ₹4.1, OMC margin ₹3.5. All figures ₹ crore per year:

Blend	Extra bn L	Consumer pays	Central excise	State VAT	Dealer comm.	OMC margin
E20	2.16	22,703	4,303	4,216	886	757
E25	3.02	31,712	6,010	5,889	1,238	1,057
E30	3.91	41,011	7,773	7,616	1,601	1,367
IB16*	1.49	15,696	2,975	2,915	613	523
IB24*	2.22	23,295	4,415	4,326	910	776

*Isobutanol rows are energy-basis (no Indian fleet-trial data). The OMC-margin column reconciles exactly with the OMC retail-profitability model (₹757 / 1,057 / 1,367 crore).

4a. The E20 → E30 walk: what each next step adds

India is already at ~E20, so the live policy question is the increment. Deltas below are each blend's take beyond what E20 already extracts:

Step (vs E20)	Δ bn L	Δ Consumer	Δ Excise	Δ VAT	Δ Dealer	Δ OMC
E25	+0.86	+9,009	+1,707	+1,673	+352	+301
E27	+1.30	+13,621	+2,581	+2,530	+532	+454
E30	+1.75	+18,308	+3,470	+3,400	+715	+610

The walk from E20 to E30 roughly doubles the volume-effect take. The increments are also qualitatively worse than the first 20%: today's engines are calibrated for E20, so the octane-recovery offset that softened E20's

real-world drop has little left to give at E25–E30 — each further step passes through closer to the raw energy loss.

4b. Diesel pool: B7 → B20 biodiesel walk

Diesel is the bigger prize by volume — 110 bn L/yr (91.4 MMT), twice the petrol pool. FAME dilutes far less per litre, but the base is enormous. Per-litre stakes: pump ₹92, excise ₹15.80, effective VAT ≈₹12.3, dealer ₹3.1, OMC ₹2.5:

Blend	Mileage drop	Extra bn L	Consumer pays	Excise	VAT	Dealer	OMC
B7	−0.6%	0.64	5,904	1,014	786	199	160
B10	−0.8%	0.92	8,455	1,452	1,126	285	230
B15	−1.2%	1.38	12,736	2,187	1,696	429	346
B20	−1.7%	1.85	17,053	2,929	2,271	575	463

B20 pulls nearly as many extra litres as E20 does on petrol, though the per-km burden is far gentler (−1.7% vs −4%). The incidence differs too: diesel is freight-dominated, so the extra cost cascades into logistics and inflation rather than household fuel budgets. Feasibility check: actual biodiesel blending was under 1% in FY24–25 and the National Biofuel Policy targets only B5 by 2030 — B15/B20 are a what-if ceiling, and OEM warranties beyond B7 remain unresolved.

5. The CBG contrast: same decarbonisation, no hidden levy

- Petrol + ethanol — energy dilution is real; pump ₹/L unchanged; consumer buys ~4% more litres at E20; centre, state, dealer and OMC all collect per-litre on the extra volume.
- Diesel + FAME — the same mechanism, mild per litre but material in aggregate because the pool is 2× petrol's.
- CNG + CBG — retailed per kilogram at methane-grade energy (46.5 MJ/kg spec floor vs 47.5 pipeline; pure methane 50). Substituting CBG changes sourcing — domestic agricultural and municipal waste instead of imported LNG — without changing the kilograms bought per kilometre. Zero dilution, zero volumetric pass-through.

Both routes substitute imports. Only the liquid route monetises an energy haircut through per-litre levies. CBG delivers the import substitution without taxing the consumer through the fuel gauge — and its SATAT procurement pricing keeps the subsidy explicit and on-budget rather than hidden in mileage.

6. CBG economics under SATAT pricing

Anchoring mileage on the SIAM/ARAI BS-VI fuel-efficiency declarations (303 models): petrol 16.67 kmpl, diesel 17.91 kmpl, CNG 27.40 km/kg — and, decisively, the same nameplate travels +40.3% further per unit on CNG than on petrol (11 factory petrol/CNG pairs).

6a. What a megajoule costs to procure

Source	₹/MJ	Basis
Petrol (refinery / trade parity)	1.81	₹58/L
Ethanol (OMC procurement)	2.94	₹62/L
Domestic APM gas	0.51	~₹24/kg CNG-eq
Imported spot RLNG	0.94	~₹45/kg CNG-eq (\$12/MMBtu)
CBG (SATAT assured)	1.16	₹54/kg ex-plant + 5% GST

Ethanol's renewable premium over the petrol it displaces is +63% per MJ — before the mileage-dilution levy on the consumer. CBG's premium over the spot RLNG it displaces is +23% per MJ, and it undercuts imported gas outright whenever spot LNG runs above ~\$14.8/MMBtu. Per megajoule of renewable energy bought, SATAT CBG (₹1.16) is 2.5× cheaper than ethanol (₹2.94), with no volumetric side effects.

6b. Cost per km on SIAM/ARAI declared FE (Delhi prices)

Fuel	₹/km	Mileage basis
Petrol E0	6.30	16.67 kmpl (SIAM declared)
Petrol E20	6.56	E20 real-world -4%
Petrol E30	6.77	E30 real-world -7%
Diesel B0	5.14	17.91 kmpl (SIAM declared)
Diesel B20	5.23	B20 energy basis -1.7%
CNG	2.74	27.40 km/kg (SIAM declared)
CBG (any % in CNG)	2.80	energy-scaled 46.5 / 47.5 MJ/kg

CNG runs at ~43% of petrol's cost per km — the +40.3% distance uplift compounding the per-unit price gap. The blend walk moves petrol the wrong way (₹6.30 → ₹6.77/km from E0 to E30) while CBG blending leaves the CNG ₹/km essentially untouched: even 100% CBG at the IS 16087 purity floor costs 2.1% in mileage, and at obligation-level shares it rounds to zero.

6c. The CBG Blending Obligation: national cost, zero consumer levy

CBO mandates 1% CBG in CGD gas from FY25-26, stepping toward ~5% by FY28-29. On the 6.67-MMT CNG(T) pool, at SATAT ₹54/kg (+5% GST) versus spot RLNG ~₹45/kg as the displaced marginal molecule:

CBO share	CBG needed (t/yr)	Procurement (₹ cr/yr)	Pump-price delta	Mileage delta
1% (FY25-26)	66,700	378	+₹0.12/kg	-0.02%
5% (FY28-29)	333,500	1,891	+₹0.59/kg	-0.11%

Even the full 5% obligation moves the CNG pump by ~₹0.59/kg (~0.8%) and mileage by ~0.1% — against E20's silent 4% mileage cut and ₹22,700 crore/yr volume dividend. Where CBG does need support, the subsidy sits on-budget and ex-plant — SATAT price assurance, GOBARDhan capex grants, and fermented-organic-manure offtake under FCO Schedule VIII — rather than hidden in the fuel gauge. Caveat: ₹54/kg is an assured floor, not proof of plant viability; feedstock economics (press-mud cheapest, agri-residue dearest) still decide whether plants get built.

7. Price-adjustment scenarios: what would neutralise the mileage loss

If blended petrol were priced for honest cost-per-km, its pump price would carry a discount equal to the mileage it takes away: $P(\text{blend}) = P(\text{E0}) \times (1 - \text{drop})$. At a ₹105 pump, that means:

Blend	Mileage drop	Parity discount	Parity pump price	National cost (₹ cr/yr)
E20	4.0%	₹4.20/L	₹100.80	22,703
E25	5.5%	₹5.78/L	₹99.22	31,712
E30	7.0%	₹7.35/L	₹97.65	41,011

Note the FY23 budget's ₹2/L penalty on unblended petrol is a stick pointing the other way — it widens the per-km gap instead of closing it. Four funding routes, tested against the ethanol-procurement, grain-price and taxation numbers from the companion models:

S1 — pass through the tax break already embedded (works, 1.3–1.5× coverage)

From the Bang-for-Your-Buck tax analysis: the ethanol molecule escapes both central excise (₹19.90/L) and state VAT (~₹19.5/L), bearing only 5% GST (~₹3.1/L on ₹62 ethanol) — but because E20 sells at the unblended price, that saving is retained in the price build-up rather than rebated. Netting off ethanol's cost premium over refinery petrol (₹62 vs ₹58):

Blend	Embedded headroom (₹/L)	Parity needs (₹/L)	Coverage
E20	6.46	4.20	1.54×
E25	8.07	5.78	1.40×
E30	9.69	7.35	1.32×

The embedded tax break more than funds parity at every blend level. Passing it through prices E20 honestly with ~₹2/L still left in the chain — no new subsidy and no new revenue loss versus an unblended counterfactual. What it does end is the windfall from the volume effect: that gain is, litre for litre, the consumer's loss.

S2 / S3 — centre or states fund it via duty cuts (possible, costly)

An excise cut of the full parity discount leaves ₹15.70/L (E20) falling to ₹12.55/L (E30) and costs the centre ₹22,700–41,000 crore/yr; the state-VAT route is symmetrical. Either is politically the mirror image of today's arrangement — which is precisely why neither has happened.

S4 — fund it from cheaper ethanol (fails on grain economics)

Parity funded from procurement alone needs ethanol at ₹41/L (E20) down to ₹37.5/L (E30) — far below every ESY 2024-25 feedstock slab:

Feedstock	OMC procurement (₹/L)
C-heavy molasses	57.97
FCI surplus rice	58.50
B-heavy molasses	60.73
Damaged food grains	64.00
Sugarcane juice / syrup	65.61
Maize	71.86

Grain and cane mandi prices set these floors — and maize, the marginal feedstock the E30 walk leans on, is the most expensive slab, pushing average procurement up, not down. Squeezing procurement to the cheapest slab funds only ₹0.81/L of E20's ₹4.20 need (~19%). Ethanol economics cannot pay for parity; only the tax side can. CBG needs no such scenario at all — its energy parity per kilogram is physical, not fiscal.

8. Caveats and sources

- Vehicle mileage: SIAM/ARAI BS-VI FE declarations (form 2344, 303 four-wheeler models, Apr 2020) from the vehicle_fuel_mileage repo — type-approval figures, used for relative fuel-type gaps; the +40.3% petrol→CNG uplift is anchored on 11 same-nameplate pairs.
- CBG/SATAT: MoPNG assured ex-plant price ₹54/kg (+5% GST); CBO 1% of CGD gas FY25-26 → ~5% FY28-29; gas comparators APM \$6.5/MMBtu, spot RLNG \$12/MMBtu at ₹83/\$.

- Volumes: PPAC Ready Reckoner FY2025-26 (H1), Table 6.1 (MS 40.0 MMT, HSD 91.4 MMT FY24-25); outlets Table 6.6/6.7 (99,281); dealer commission Table 8.10.
- Mileage drops: SIAM/ARAI central figures (E20 4%, band 2–6%; up to ~12% for older vehicles); E25/E30 scaled on ethanol's calorific deficit; isobutanol and biodiesel rows are energy-basis.
- Per-litre levies are editable levers shared with `omc_model.py` and `statewise_tax_impact.py`: excise ₹19.90/L petrol, ₹15.80/L diesel; VAT 25% on ₹78/L petrol base, 17.5% on ₹70/L diesel base; OMC margins ₹3.5/₹2.5.
- This volume effect is distinct from — and additive to — the tax-differential effect (states forgo VAT on the ethanol fraction, ~₹9,000 cr/yr at E20): states lose on substitution while every per-litre stakeholder gains on volume.
- Generated by `build_energy_doc.js` from `energy_blend_comparison.py` outputs; do not hand-edit.
Repository: github.com/herrickshaw/india-omc-fuel-fleet-model.